PART 1: Data Understanding

1. How many datasets were provided?

The ShopSmart Analysis dataset contains nine tables(within one dataset) containing information about customers, products, orders, stores, returns, employees, suppliers, inventory, and order items.

2. What does each dataset represent?

- Customer: contains customers information, including names, gender, age, contact details, location, registration date, & customer segment.
- Product: contains product information, including product names, categories, brands, cost prices, & selling prices.
- Orders: contains sales transaction information, including customers, products, stores, order dates, quantities, prices, & discount.
- Stores: contains information about store location, names, regions, states, cities, & opening date.
- Returns: Records returned orders, including return date, reasons, refund amounts, & return rates
- Employees: contains employee details, including names, roles, store assignments, hire date, & monthly salaries.
- Suppliers: contains supplier contact information and locations, including supplier names, emails, phone numbers, cities, & countries.
- Inventory: Track product stock levels at each store, including stock on hand, reorder levels, & last restock dates.
- Order items: contains order line details, including products, quantites, unit price, discounts, and line totals.

The tables cover different aspects of ShopSmart's retail operation, allowing analysis of sales performance, customer behaviour, product performance, store operations, inventory management, & product returns.

3. Identify the Primary Key of each table.

A Primary Key (PK) is a column, or combination of columns, that uniquely identifies each record in a table. A primary key cannot contain duplicate values or blank values.

- Customers: customer_id
- Products: product_id

- Orders: order-id
- Stores: store_id
- Returns: return_id
- Employees: employee_id
- Suppliers: supplier_id
- Inventory: inventory_id
- Order_items: order_item_id

4. Identify possible Foreign Keys.

A Foreign Key (FK) is a column that refers to a primary key in another table. Foreign keys allow tables to be connected.

- Orders: customer_id
- Orders: store_id,
- Orders: product_id
- Returns: order_id
- Employees: store_id
- Inventory: product_id
- Inventory: store_id
- Order items: product_id
- Order items: order_id

5. Draw the relationship between the tables.

